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trans-RhCl(CO)(PPh_3)$_2$-Catalyzed Monomeric and Dimeric Cycloisomerization of Propargylic 2,3-Dienoates. Establishment of α,β-Unsaturated δ-Lactone Rings by Cyclometallation

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Abstract: Cyclometallation of two unsaturated carbon−carbon bonds usually requires the application of low-valent metal catalysts, which could cleave the propargylic ester linkage. Thus, it is desirable to identify a catalyst which could undergo cyclometallation without cleaving the propargylic ester linkage. In this paper, we used *trans*-RhCl(CO)(PPh_3)$_2$ to realize the cyclometallation of propargylic 2,3-dienoates. The substituents at the 4-position of allenoate moiety nicely control the reaction pathway: when the 4-position of propargylic 2,3-dienoate **1** was monosubstituted with an aryl group, the bicyclic intermediate **7** formed by the cyclometallation could highly selectively undergo carbometalation with the alkyne moiety in the second molecule of propargylic 2,3-dienoate **1** to afford metallabicyclic intermediates **8a** or **8b**. Subsequent reductive elimination would afford **9**, which could undergo an intramolecular Diels−Alder reaction resulting in the formation of polycyclic bis(δ-lactone)-containing structures **2**. The intermediate could be trapped by adding 3-methoxyprop-1-yne affording cyclization−aromatization product **4p** highly selectively. If the substituent at the 4-positon of the 2,3-allenoate moiety has a β-H atom, sequential unimolecular cyclometallation/β-H elimination/reductive elimination occurs to afford cross-conjugated 5(*Z*)-alkylidene-4-alkenyl-5,6-dihydropyran-2-ones. The *Z*-stereochemistry of the exo double bond was determined by the cyclometallation. Some of the α,β-unsaturated δ-lactones could be easily converted to other synthetically useful compounds via reduction reaction, hydrogenation, and iodination/coupling protocol.

Introduction

Cyclometallation involving two unsaturated carbon−carbon bonds has been one of the hottest topics in organometallic chemistry.[1] Recently, we observed a bimolecular cyclization of unsubstituted 1,5-bisallenes leading to the formation of 18,19-norsteroid derivatives with the help of a catalytic amount of *trans*-RhCl(CO)(PPh_3)$_2$;[2] under the catalysis of Pd(0), two internal C═C bonds in unsubstituted 1,5-bisallenes could undergo cyclometallation[3] (Scheme 1).

Brummond[4] and Shibata[5] have pioneered the cyclization of allene-ynes for the synthesis of six-membered carbocycles, heterocycles, and δ,ϵ-lactams. However, the synthesis of lactones[6] has rarely been reported probably due to the possible cleavage of the ester linkage. In this paper, we wish to report our own observation that propargylic 2,3-dienoates could undergo a bimolecular cyclization[2] or cyclometalation/β-H elimination/reductive elimination sequence[4,5] to afford α,β-unsaturated δ-lactones depending on the substituents on the 4-position of allene moieties (Scheme 2).

Results and Discussion

Bimolecular Cyclization of Propargyl 4-Aryl-2,3-alkadienoates 1a−1e.[7] The reaction of compound **1a** under the catalysis of 5 mol % *trans*-RhCl(CO)(PPh_3)$_2$ in toluene[2] afforded a pair of diastereoisomers **2a** and *epi*-**2a**, which could be easily separated by flash chromatography on silica gel. The structures of these dimeric cyclization products were unambiguously established by the X-ray diffraction study (Figure 1).[8] It should be noted that although there are four stereogenic centers in the products, only two diastereoisomers were formed.

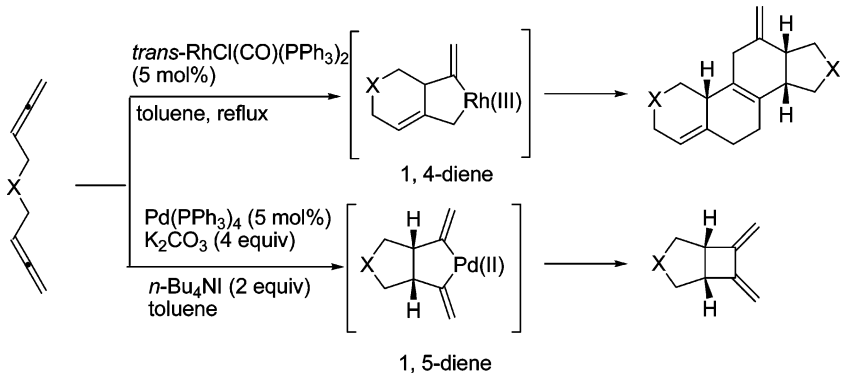
This bimolecular cyclization reaction of propargylic 2,3-dienoates is very sensitive to the concentration of the reaction (Table 1).

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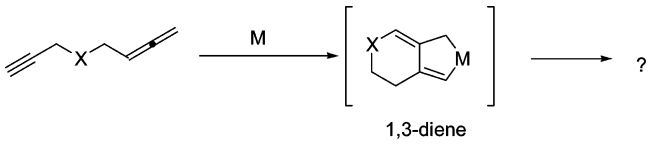
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Scheme 1. Cyclometallation of 1,5-Bisallenes

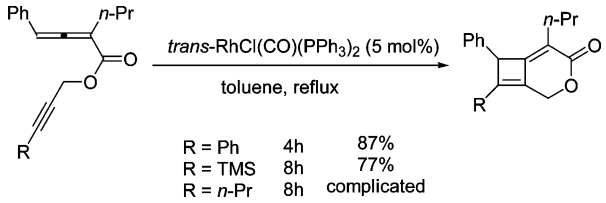


Scheme 2. Cyclometallation of 1,5-Allenynes



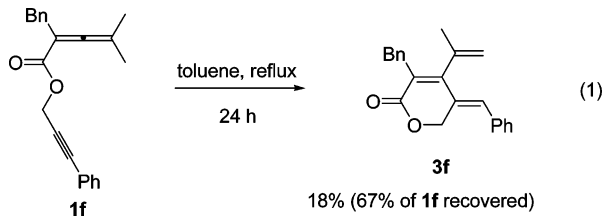
We obtained the highest yields of product **2a** when the reaction was carried out in xylene (0.042 M) at 140 °C in a reaction tube with a screw cap. The high temperature is believed to be required by the Diels−Alder reaction. With the optimized conditions in hand, we studied the scope of this reaction (Table 2). Preliminary results showed that Ar can be a Ph- (Table 2, entries 1−4) and *p*-MeO-substituted phenyl group (Table 2, entry 5); R could be a benzyl group (Table 2, entry 1) and a primary or secondary alkyl group (Table 2, entries 2−5). The low-yielding nature observed in some cases could be caused by the high reaction temperature applied, which could lead to the polymerization of the starting materials. It should be noted that 4-aryl-2,3-dienoates possessing an internal alkyne moiety with phenyl or TMS underwent an intramolecular [2 + 2]-cycloaddition reaction instead of dimeric cyclization probably due to both the steric and electronic effects.[9] With R being *n*-propyl, the reaction is complicated (Scheme 3).

Scheme 3. *trans*-RhCl(CO)(PPh$_3$)$_2$-Catalyzed Reaction of 4-Aryl-2,3-dienoates Bearing an Internal Carbon−Carbon Triple Bond



Unimolecular Cyclization of Propargyl 4-Alkyl-2,3-alkadienoates 1f−1o. However, it is interesting to observe when two methyl groups were installed at the 4-position of the allene moiety, i.e., **1f**, (*Z*)-3-benzyl-5-benzylidene-4-(prop-1-en-2-yl)-5,6-dihydropyran-2-one **3f** (Figure 2)[11] was formed.

The control reaction shows that the ene reaction could also occur under thermal conditions, but with a lower yield and longer reaction time (eq 1).



Thus, this sequential unimolecular cyclometallation/β-H elimination/reductive elimination of propargylic 2,3-dienoates **1f−1n** was studied (Table 3). This is the first example of an Alder-ene type reaction of an ester-tethered substrate to form a six-membered lactone ring.[4,5,12] In this study, it was observed that the application of [Rh(CO)$_2$Cl]$_2$ used by the Brummond group for the cyclization of amides and ethers[4] led to the decomposition of the starting propargylic 2,3-dienoates due to the cleavage of the propargylic ester linkage.[4,13] The reaction

(8) Crystal data for compound **2a**: $C_{40}H_{32}O_4$, MW = 576.66, orthorhombic, space group *Pbca*, final *R* indices [$I > 2\sigma(I)$], R1 = 0.0506, wR2 = 0.1099, *R* indices (all data), R1 = 0.0999, wR2 = 0.1265, *a* = 24.367 (2) Å, *b* = 8.1958 (7) Å, *c* = 30.325 (3) Å, α = 90°, β = 90°, γ = 90°, *V* = 6056.2-(9) Å^3, *T* = 293 (2) K, *Z* = 8, reflections collected/unique: 33 787/6614 (*R*(int) = 0.1050). Supplementary crystallographic data have been deposited at the Cambridge Crystallographic Data Center. CCDC: 632768. Crystal data for compound *epi*-**2a**: $C_{40}H_{32}O_4$, MW = 576.66, monoclinic, space group *P*2(1)/*c*, final *R* indices [$I > 2\sigma(I)$], R1 = 0.0854, wR2 = 0.1591, *R* indices (all data), R1 = 0.2057, wR2 = 0.2011, *a* = 11.4483 (13) Å, *b* = 30.370 (3) Å, *c* = 8.4474 (10) Å, α = 90°, β = 98.854(3)°, γ = 90°, *V* = 2902.0(6) Å^3, *T* = 293 (2) K, *Z* = 4, reflections collected/unique: 15 200/5383 (*R*(int) = 0.1430). Supplementary crystallographic data have been deposited at the Cambridge Crystallographic Data Center. CCDC: 632770.

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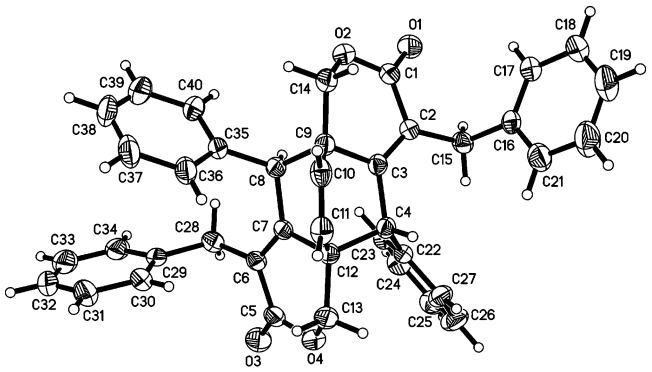
(10) Crystal data for compound **2d**: $C_{34}H_{36}O_4$, MW = 508.63, triclinic, space group *P*1, final *R* indices [$I > 2\sigma(I)$], R1 = 0.0900, wR2 = 0.2104, *R* indices (all data), R1 = 0.2184, wR2 = 0.2722, *a* = 8.6809 (14) Å, *b* = 15.504 (2) Å, *c* = 21.838 (3) Å, α = 93.347°, β = 92.725°, γ = 94.441°, *V* = 2921.4(8) Å^3, *T* = 293 (2) K, *Z* = 4, reflections collected/unique: 15 521/10707 (*R*(int) = 0.1336). Supplementary crystallographic data have been deposited at the Cambridge Crystallographic Data Center. CCDC: 647587. Crystal data for compound *epi*-**2d**: $C_{34}H_{36}O_4$, MW = 508.63, orthorhombic, space group *Pca*2(1), final *R* indices [$I > 2\sigma(I)$], R1 = 0.0765, wR2 = 0.1711, *R* indices (all data), R1 = 0.1616, wR2 = 0.2112, *a* = 28.980 (8) Å, *b* = 9.097 (2) Å, *c* = 21.175 (5) Å, α = 90°, β = 90°, γ = 90°, *V* = 5582(3) Å^3, *T* = 293 (2) K, *Z* = 8, reflections collected/unique: 29 213/5646 (*R*(int) = 0.1745). Supplementary crystallographic data have been deposited at the Cambridge Crystallographic Data Center. CCDC: 647588.

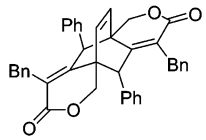
(11) Crystal data for compound **3f**: $C_{22}H_{20}O_2$, MW = 316.38, orthorhombic, space group *P*2(1)2(1)2(1), final *R* indices [$I > 2\sigma(I)$], R1 = 0.0592, wR2 = 0.1473, *R* indices (all data), R1 = 0.0629, wR2 = 0.1500, *a* = 6.0205 (7) Å, *b* = 16.6705 (18) Å, *c* = 17.5656 (19) Å, α = 90°, β = 90°, γ = 90°, *V* = 1763.0(3) Å^3, *T* = 293 (2) K, *Z* = 4, reflections collected/unique: 10 498/2238 (*R*(int) = 0.1078). Supplementary crystallographic data have been deposited at the Cambridge Crystallographic Data Center. CCDC: 632769.

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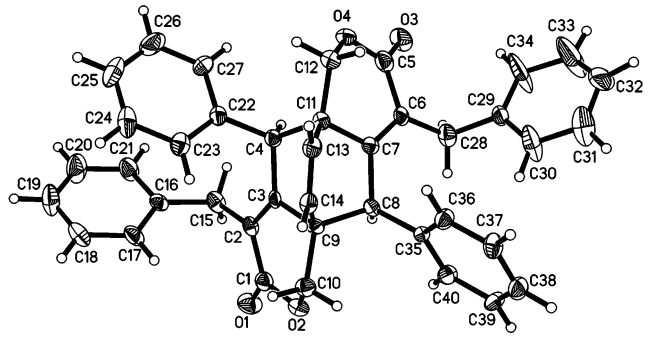
(13) For Rh(I)-catalyzed cleaving C−O bond of allylic esters, see: Evans, P. A. *Modern Rhodium-Catalyzed Organic Reactions*; Wiley-VCH: 2005; pp 191−214. For Pd(0)-catalyzed cleaving C−O bond of propargyl carbonates, see: Tsuji, J. *Palladium Reagents and Catalysts*; Wiley: Chichester, 2004; pp 543−563.

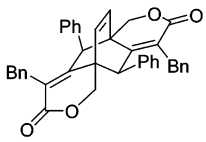
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2a



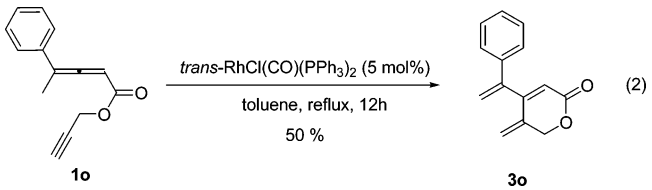


epi-**2a**

Figure 1. ORTEP representation of **2a** and *epi*-**2a** with thermal ellipsoids at the 30% probability level.

tolerates different substituents at the alkyne moiety (Ph, TMS, Bu, H, Table 3, entries 1−9) and alkyl substiuents at the allene, which could afford the α,β-unsaturated δ-lactones **3** in moderate to good yields. The exo C═C bond in **3** is in a *Z*-configuration.

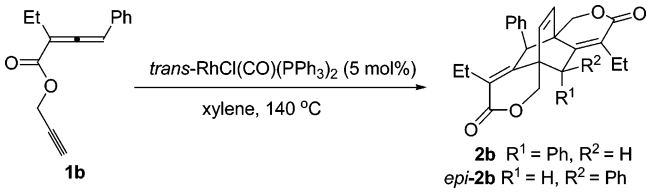
When both phenyl and methyl groups were introduced at the 4-position of the allene moiety, i.e., **1o**, 5-methylene-4-(1-phenylvinyl)-5,6-dihydropyran-2-one **3o** was formed, implying that the β-hydride elimination is still favored even with only one alkyl group bearing hydrogens to be eliminated (eq 2).



1o **3o** (2)

Mechanistic Study. In order to trap any intermediate formed in this reacion, the cyclization of **1p** (an equivalent of **1c**) was conducted in the presence of 3-methoxyprop-1-yne. The reaction afforded 5-(4-bromophenyl)-6-(methoxymethyl)-4-propylisochroman-3-one **4p** in 50% yield (Scheme 4). The structure of **4p**

Table 1. Effect of Concentration of **1b** in Xylene on the *trans*-RhCl(CO)$(PPh_3)_2$-Catalyzed Dimeric Cyclization of Propargyl 2-Ethyl-4-phenylbuta-2,3-dienoate **1b**[a]



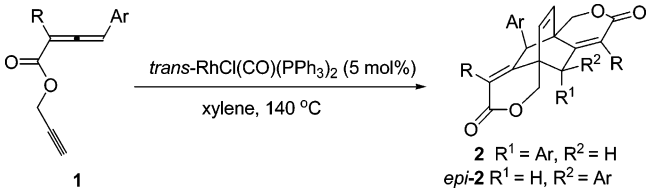
1b

2b R^1 = Ph, R^2 = H
epi-**2b** R^1 = H, R^2 = Ph

entry	concentration of **1b** (M)	isolated yield of **2b** (%)	isolated yield of *epi*-**2b** (%)
1	0.0833	30	18
2	0.0417	52	23
3	0.0278	38	22

[a] The reaction was conducted at 140 °C in a Schlenk tube with a screw cap.

Table 2. Dimeric Cyclization of Propargyl 2,3-Dienoates[a]



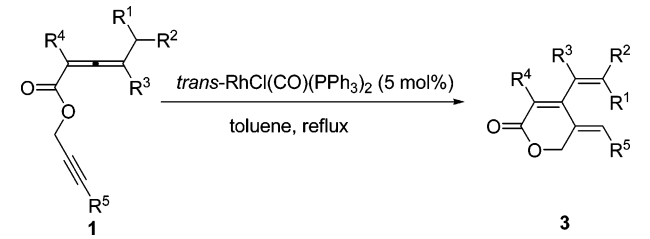
1

2 R^1 = Ar, R^2 = H
epi-**2** R^1 = H, R^2 = Ar

	1					
entry	Ar	R		time (h)	isolated yield of **2** (%)	isolated yield of *epi*-**2** (%)
1	Ph	Bn	**(1a)**	10	27 (**2a**)[8]	13 (*epi*-**2a**)[8]
2	Ph	Et	**(1b)**	12	52 (**2b**)	23 (*epi*-**2b**)
3	Ph	*n*-Pr	**(1c)**	10	45 (**2c**)	21 (*epi*-**2c**)
4	Ph	*i*-Bu	**(1d)**	10	38 (**2d**)[10]	28 (*epi*-**2d**)[10]
5	*p*-$MeOC_6H_4$	*n*-Bu	**(1e)**	10	45 (**2e**)	25 (*epi*-**2e**)

[a] The reaction was conducted at 140 °C in a Schlenk tube with a screw cap using 5 mol % *trans*-RhCl(CO)$(PPh_3)_2$ at a concentration of 0.042 M in xylene.

Table 3. Unimolecular Cyclization of Propargylic 2,3-Alkadienoates **1f**−**1n**



1 **3**

	1							
entry	R^1	R^2	R^3	R^4	R^5		time (h)	isolated yield of **3** (%)
1	H	H	Me	Bn	Ph	**(1f)**	15	67 (**3f**)
2	−$(CH_2)_5$−		H	H	TMS	**(1g)**	8	78 (**3g**)
3	H	H	Me	Bn	TMS	**(1h)**	34	66 (**3h**)
4	H	H	Me	Bn	*n*-Bu	**(1i)**	48	84 (**3i**)
5	−$(CH_2)_5$−		H	H	*n*-Bu	**(1j)**	27	63 (**3j**)
6	H	H	Me	Me	*n*-Bu	**(1k)**	34	58 (**3k**)
7	H	H	Me	Me	H	**(1l)**	17	75 (**3l**)
8	H	H	Me	Me	TMS	**(1m)**	24	53(**3m**)
9	H	H	Me	H	Ph	**(1n)**	14	63(**3n**)

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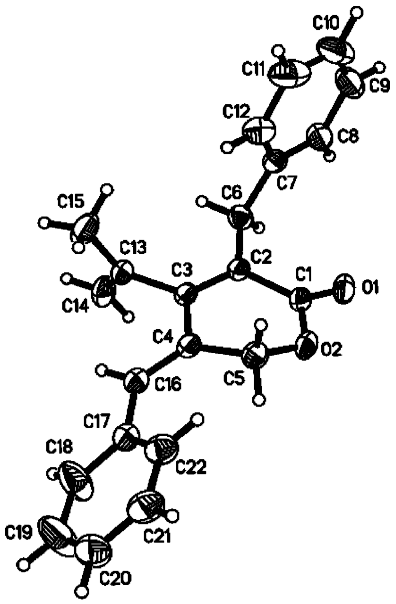
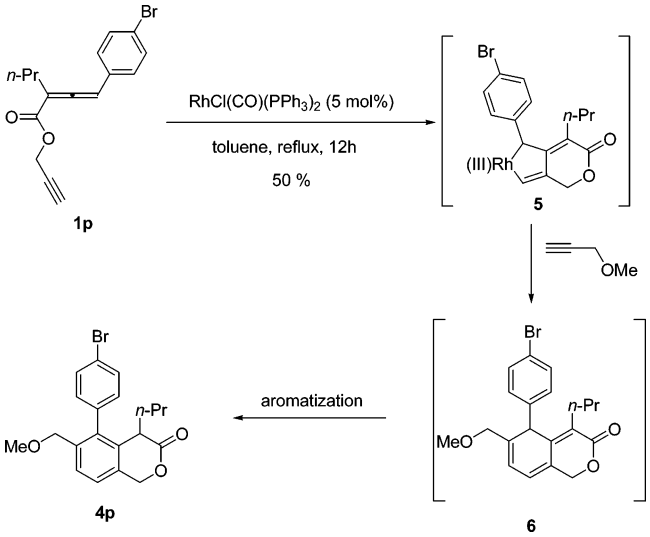


Figure 2. ORTEP representation of **3f** with thermal ellipsoids at the 30% probability level.

Scheme 4. Cyclization of Propargyl 2,3-Alkadienoate **1p** in the Presence of 3-Methoxyprop-1-yne



was confirmed via an X-ray diffraction study (Figure 3).[14] With the unequivocal structural information of compound **4p**, it was reasoned that this compound was formed via the aromatization of **6**, which, in turn, was produced by the insertion of the triple bond in progargyl methyl ether to the C−Rh bond in bicyclic

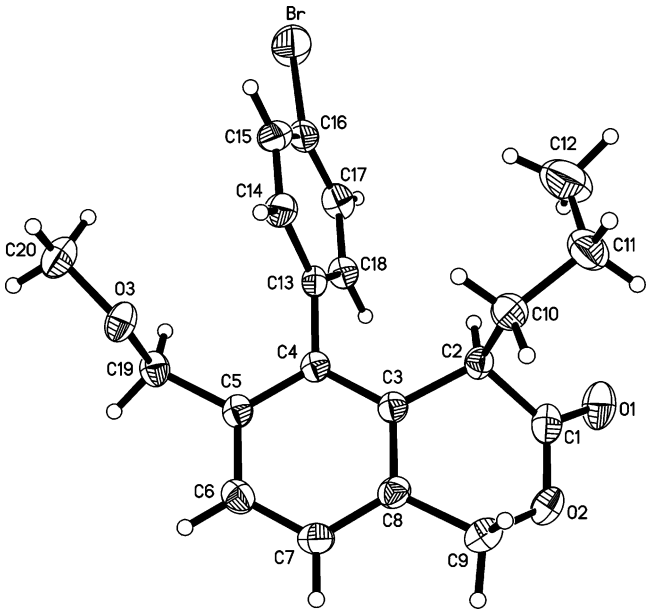


Figure 3. ORTEP representation of **4p** with thermal ellipsoids at the 30% probability level.

intermediate **5** and subsequent reductive elimination. This provided strong evidence for the mechanism of this reaction.

Based on the above results, a possible mechanistic rationale was proposed for the conversion of **1** to **2**/*epi*-**2** and **3** (Scheme 5). The reaction could proceed via the cyclometallation[2,4,5,15] of the carbon−carbon triple bond and the remote double bond of the tethered allene in **1** affording metallabicyclic intermediate **7**; if the substituent at the 4-positon of the dienoate moiety has a β-H atom, the β-H elimination occurs to form rhodium hydride intermediate **10**. Subsequent reductive elimination afforded 5,6-dihydropyran-2-one product **3**. The *Z*-stereochemistry of the exo double bond was determined by the cyclometallation and stereoselective reductive elimination. If there is no hydrogen available for β-hydride elimination by replacing an alkyl group with an aryl group, this bicyclic intermediate could highly selectively undergo carbometallation with the alkyne moiety in the second molecule of propargylic 2,3-dienoate **1** to afford metallabicyclic intermediates **8a** or **8b**. Subsequent reductive elimination would afford **9**, which could undergo an intramolecular Diels−Alder reaction[2,15a,16,17] to form the bridged polycyclic product **2** and its epimer *epi*-**2** (Scheme 5).

Derivatization of α,β-Unsaturated δ-Lactone. α,β-Unsaturated δ-lactones are widely distributed in both plants and fungi and possess a diverse range of biological activity. They have

(14) Crystal data for compound **4p**: $C_{20}H_{21}BrO_3$, MW = 389.28, monoclinic, space group *P*2(1)/*c*, final *R* indices [$I > 2\sigma(I)$], R1 = 0.0581, wR2 = 0.1345, *R* indices (all data), R1 = 0.1034, wR2 = 0.1637, *a* = 15.432 (2) Å, *b* = 14.140 (2) Å, *c* = 8.4535 (12) Å, α = 90°, β = 102.317(3)°, γ = 90°, *V* = 1802.1(4) Å^3, *T* = 293 (2) K, *Z* = 4, reflections collected/unique: 10 456/3915 (*R*(int) = 0.1458). Supplementary crystallographic data have been deposited at the Cambridge Crystallographic Data Center. CCDC: 647589.

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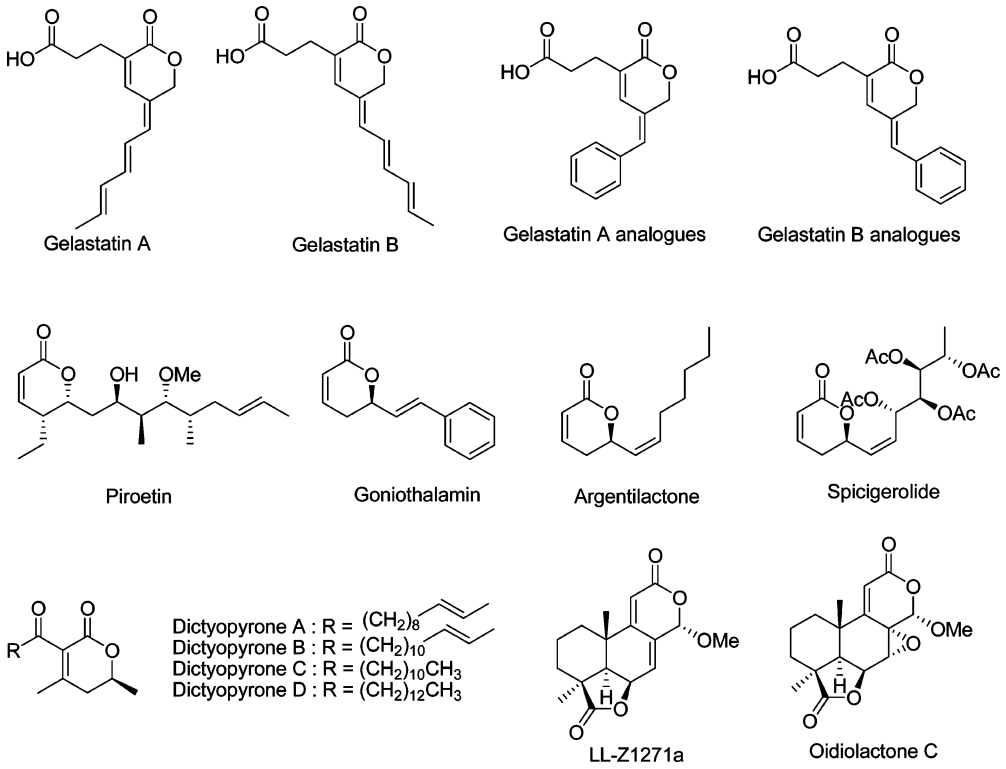


Figure 4. Some examples of natural α,β-unsaturated δ-lactones.

been reported as plant growth inhibitors, insect antifeedants, and antifungal/antitumor agents.[18] For example, gelastatin A, B and their analogues appear to be a viable lead in searching for therapeutically useful MMP inhibitors;[19] pironetin has been found to inhibit the cell cycle progression in the M phase;[20] goniothalamin has been shown to induce the apoptotic process;[21] argentilactone is an anticancer agent;[22] spicigerolide exhibits cytotoxic activity;[23] α-Pyronoids (dictyopyrones A−D) have been utilized as *anti*-leukemic agents;[24] LL-Z1271α and oidiolactone C demonstrate cytotoxic and antimicrobial activity.[25] These compounds indicate the importance of this methodology.

On the other hand, this type of products may also be used as important intermediates in organic synthesis. The carbonyl group in β-vinyl-γ-benzylidene-α,β-unsaturated δ-lactone **3f** could be reduced to a hydroxy group affording hemiacetal **11f** upon its treatment with DIBAL-H; The relatively electron-rich vinylic and exocyclic C=C bonds in **3f** could be easily hydrogenated with high regioselectivity to afford α,β-unsaturated δ-lactone **12f** (Scheme 6).

The vinylic silane moiety in product **3l** could be converted to vinylic iodide **13m** in 88% yield by its treatment with NIS. This compound could be used for subsequent Suzuki coupling and Sonogashira coupling reaction to afford **14m**, **15m**, and **16m** (Scheme 7).

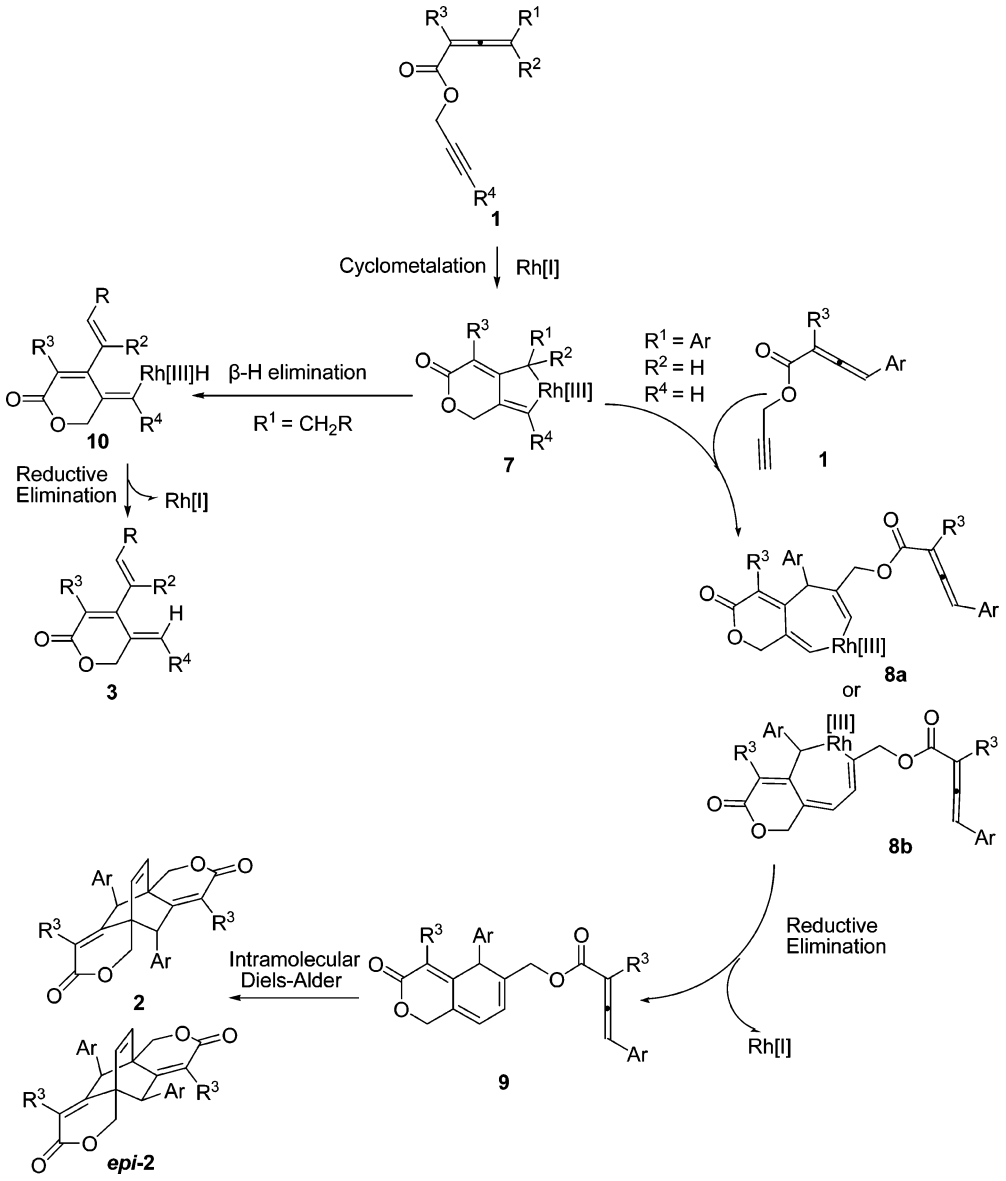
Conclusion

In conclusion, both monomeric and dimeric cyclization of propargylic 2,3-dienoates were demonstrated for the highly selective synthesis of δ-lactones. The substituents at the 4-position of 2,3-alkadienoate moiety cleanly control the reaction pathway: when the 4-position was substituted with an aryl group, bimolecular cyclization occurred resulting in the formation of polycyclic bis(δ-lactone)-containing structures; with an alkyl substitutent at this position, unimolecular cyclization occurred, leading to cross-conjugated 5(*Z*)-alkylidene-4-alkenyl-5,6-dihydropyran-2-ones. It should be noted that the propargylic 2,3-allenoic acid ester linkage survived under the current protocol. Due to the importance of α,β-unsaturated δ-lactone,[26] this method could be potentially useful in organic synthesis and medicinal chemistry. Further studies in this area are being pursued in our laboratory.

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Scheme 5. Proposed Mechanism for the Mono- or Dimeric Cyclizations of Propargylic 2,3-Alkadienoates Catalyzed by *trans*-RhCl(CO)(PPh_3)$_2$



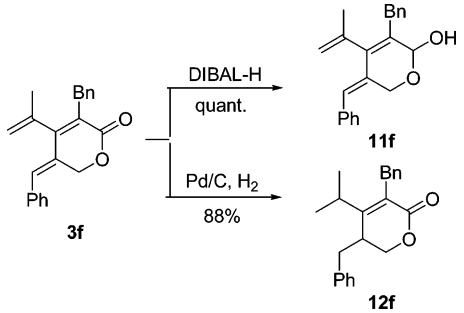
Experimental Section

General Procedure I. A solution of 2,3-dienoate acid,[7] propargylic bromide, and K_2CO_3 in acetone was stirred at room temperature. After the reaction was complete as monitored by TLC (petroleum ether/ethyl acetate = 10:1), filtration, rotary evaporation, and flash chromatography on silica gel (eluent: petroleum ether/ethyl ether = 30:1) afforded **1a**−**p**.

Preparation of Prop-2′-ynyl 2-Benzyl-4-phenylbuta-2,3-dienoate (1a). A solution of 2-benzyl-4-phenylbuta-2,3-dienoic acid (1.003 g, 4 mmol), propargyl bromide (1.6 g, 13 mmol), and K_2CO_3 (704 mg, 5 mmol) in acetone (40 mL) was stirred at room temperature for 12 h to afford 859 mg (74%) of **1a**: Liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.37−7.15 (m, 10 H), 6.55 (t, J = 2.4 Hz, 1 H), 4.77 (dd, J = 15.9, 2.7 Hz, 1 H), 4.68 (dd, J = 15.9, 2.7 Hz, 1 H), 3.74 (dd, J = 14.7, 2.1 Hz, 1 H), 3.68 (dd, J = 14.7, 2.1 Hz, 1 H), 2.43 (t, J = 2.1 Hz, 1 H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 213.5, 165.6, 138.7, 131.8, 128.9, 128.8, 128.3, 127.9, 127.4, 126.5, 103.6, 99.0, 77.6, 74.9, 52.6, 35.5; MS (EI) *m*/*z* (%) 288 (M^+, 1.02), 91 (100); IR (neat) 3288, 3030, 2120, 1944, 1718, 1600, 1496, 1454, 1260, 1086, 1063 cm^{-1}; HRMS (EI) calcd for $C_{20}H_{16}O_2$ (M^+) 288.1150. Found 288.1134.

General Procedure II. A solution of propargylic 2,3-dienoate **1a**−**1e** and **1p** (0.25 mmol) and *trans*-RhCl(CO)(PPh_3)$_2$ (8 mg, 0.0125 mmol) in 6 mL of dry xylene was heated at 140 °C in a Schlenk tube with a screw cap. After the reaction was complete as monitored by TLC (petroleum ether/ethyl acetate = 3:1), rotary evaporation, and flash chromatography on silica gel (eluent: petroleum ether/ethyl acetate = 5:1) afforded **2a**−**e** and **2p** (more polar) and *epi*-**2a**−**e** and *epi*-**2p** (less polar).

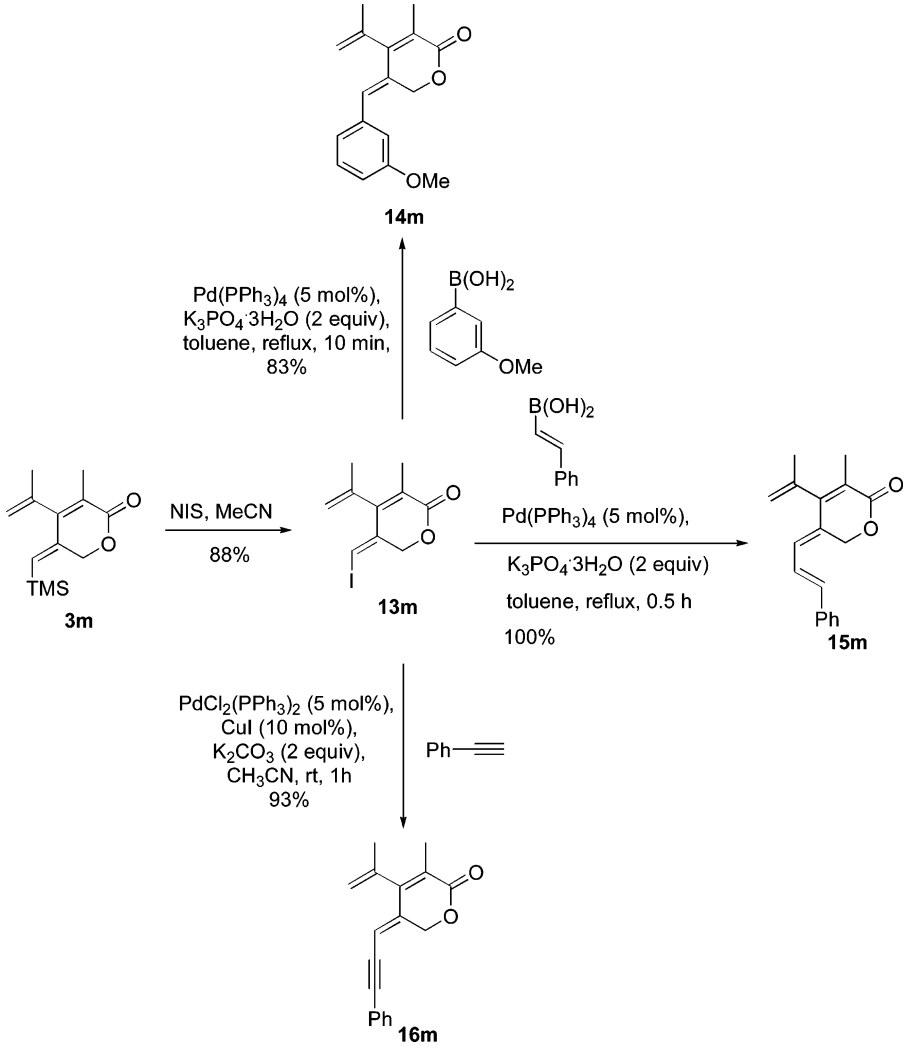
Scheme 6. Derivatization of α,β-Unsaturated δ-Lactone **3f**



Preparation of Products 2a and *epi*-2a. A solution of prop-2′-ynyl 2-benzyl-4-phenylbuta-2,3-dienoate **1a** (70 mg, 0.25 mmol) and *trans*-RhCl(CO)(PPh_3)$_2$ (9 mg, 0.0125 mmol) in 6 mL of dry xylene

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Scheme 7. Iodination of Lactone **3m** and the Coupling Reaction of the Iodide **13m**



was heated at 140 °C for 10 h to afford 19 mg (28%) of **2a** (more polar) and 9 mg (13%) of *epi*-**2a** (less polar). **2a**: Solid, mp 259−260 °C (petroleum ether/ethyl acetate); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.45−7.26 (m, 4 H), 7.26−7.00 (m, 10 H), 6.93−6.80 (m, 6 H), 6.41 (d, J = 7.5 Hz, 1 H), 5.82 (d, J = 7.5 Hz, 1 H), 4.70 (d, J = 12.3 Hz, 1 H), 4.61 (d, J = 12.3 Hz, 1 H), 4.45 (d, J = 12.0 Hz, 1 H), 4.24 (s, 1 H), 4.17 (d, J = 12.0 Hz, 1 H), 3.77 (s, 1 H), 3.48 (d, J = 14.1 Hz, 1 H), 3.40 (d, J = 15.0 Hz, 1 H), 2.99 (d, J = 14.1 Hz, 1 H), 2.73 (d, J = 15.0 Hz, 1 H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 164.8, 163.0, 156.4, 152.2, 138.6, 137.2, 136.8, 135.54, 135.48, 132.1, 130.4, 129.11, 129.07, 128.8, 128.5, 128.34, 128.32, 128.26, 128.0, 126.3, 126.1, 69.2, 67.0, 57.7, 51.1, 46.8, 46.3, 33.7, 33.4; MS (ESI) m/z (%) 615 (M + K^+, 30), 577 (M + H^+, 100); IR (neat) 1706, 1701, 1602, 1495, 1465, 1453, 1198, 1151 cm^{-1}; HRMS (ESI) calcd for $C_{40}H_{32}O_4Na$ (M + Na^+) 599.2193. Found 599.2187.

epi-**2a**: Solid, mp 274−275 °C (petroleum ether/ethyl acetate); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.35−7.25 (m, 6 H), 7.12−7.00 (m, 10 H), 6.70−6.60 (m, 4 H), 5.86 (s, 2 H), 4.52 (d, J = 12.0 Hz, 2 H), 4.43 (d, J = 12.0 Hz, 2 H), 4.17 (s, 2 H), 3.43 (d, J = 14.4 Hz, 2 H), 3.36 (d, J = 14.4 Hz, 2 H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 164.0, 157.6, 137.4, 136.5, 131.0, 130.4, 128.6, 128.2, 128.1, 126.2, 66.5, 52.8, 46.6, 34.0; MS (ESI) m/z (%) 615 (M + K^+, 25), 599 (M + Na^+, 20), 594 (M + NH_4^+, 40), 577 (M + H^+, 100); IR (neat) 1716, 1601, 1494, 1452, 1310, 1142 cm^{-1}; HRMS (ESI) calcd for $C_{40}H_{33}O_4$ (M + H^+) 577.2373. Found 577.2372.

General Procedure III: A solution of propargylic 2,3-dienoate **1f**−**1o** (0.25 mmol) and *trans*-RhCl(CO)(PPh_3)$_2$ (9 mg, 0.0125 mmol) in 6 mL of dry toluene was refluxed under Ar. After the reaction was complete as monitored by TLC (petroleum ether/ethyl acetate = 10:1), rotary evaporation and flash chromatography on silica gel (eluent: petroleum ether/ethyl ether = 20:1) afforded the product **3f**−**o.**

Synthesis of (*Z*)-3-Benzyl-5-benzylidene-4-(prop-1-en-2-yl)-5,6-dihydropyran-2-one (3f). A solution of 3′-phenylprop-2′-ynyl 2-benzyl-4-methylpenta-2,3-dienoate **1f** (64 mg, 0.20 mmol) and *trans*-RhCl(CO)(PPh_3)$_2$ (7 mg, 0.01 mmol) in 5 mL of dry toluene was refluxed under Ar for 15 h to afford 43 mg (67%) of **3f**. Some starting material remained. Solid, mp 89−90 °C (petroleum ether/ethyl acetate); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.45−7.10 (m, 10 H), 6.89 (s, 1 H), 5.41 (s, 1 H), 5.23 (s, 2 H), 5.00 (s, 1 H), 3.87 (s, 2 H), 1.94 (s, 3 H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 164.6, 153.6, 139.7, 139.4, 135.0, 133.2, 129.0, 128.6, 128.5, 128.4, 128.3, 126.0, 124.9, 118.0, 66.5, 34.0, 23.1; MS (EI) m/z (%) 316 (M^+, 0.57), 91 (100); IR (neat) 1699, 1602, 1494, 1446, 1413, 1344, 1272, 1263, 1185, 1122, 1059 cm^{-1}. Anal. Calcd for $C_{22}H_{20}O_2$: C, 83.51; H, 6.37. Found: C, 83.45; H, 6.26.

Mechanistic Study. Synthesis of 5-(*p*-Bromophenyl)-6-(methoxymethyl)-4-propylisochroman-3-one (4p). A solution of prop-2′-ynyl 2-propyl-4-(*p*-bromophenyl)buta-2,3-dienoate **1p** (46 mg, 0.15 mmol), 3-methoxyprop-1-yne (110 mg, 1.57 mmol), and *trans*-RhCl(CO)(PPh_3)$_2$ (6 mg, 0.0087 mmol) in 3.6 mL of dry toluene was heated at 110 °C in the reaction tube with a screw cap for 10 h to afford 28 mg (50%) of the product **4p**: Solid, mp 71−72 °C (petroleum ether/ethyl acetate); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.61−7.55 (m, 2 H), 7.45 (d, J = 7.8 Hz, 1 H), 7.23 (d, J = 7.8 Hz, 1 H), 7.12−7.07 (m,

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1 H), 7.07−6.98 (m, 1 H), 5.57 (d, J = 14.3 Hz, 1 H), 5.23 (d, J = 14.3 Hz, 1 H), 4.06 (s, 2 H), 3.55−3.45 (m, 1 H), 3.24 (s, 3 H), 1.80−1.60 (m, 2 H), 1.45−1.10 (m, 2 H), 0.70 (t, J = 7.2 Hz, 3 H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 172.3, 138.4, 137.1, 135.3, 133.9, 131.9, 131.4, 131.2, 131.0, 130.2, 127.1, 124.5, 122.1, 72.2, 69.5, 58.4, 43.3, 33.5, 20.3, 13.3; MS (EI) m/z (%) 390 (M^+, ^{81}Br, 5.61), 388 (M^+, ^{79}Br, 5.76), 43 (100); IR (neat) 2959, 2929, 1739, 1490, 1465, 1384, 1234, 1194, 1096 cm^{-1}; HRMS (EI) calcd for $C_{20}H_{21}O_3Br$ (M^+) 388.0674. Found 388.0664.

Synthetic Application. 1. Synthesis of (*Z*)-3-Benzyl-5-benzylidene-4-(prop-1-en-2-yl)-5,6-dihydro-2*H*-pyran-2-ol (11f). To a solution of (*Z*)-3-benzyl-5-benzylidene-4-(prop-1-en-2-yl)-5,6-dihydro-pyran-2-one **3f** (50 mg, 0.16 mmol) in 1 mL of dry THF at −78 °C was added DIBAL-H (1 M in toluene, 0.64 mL) for 5 min. After quenching the reaction with saturated NaCl, extraction with ether, drying over Na_2SO_4, filtration, and evaporation, the residue was purified by column chromatography on silica gel to afford 55 mg (100%) of the product **11f**: Liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.40−7.12 (m, 10 H), 6.63 (s, 1 H), 5.35 (s, 1 H), 5.19 (d, J = 4.8 Hz, 1 H), 5.04 (s, 1 H), 4.87 (d, J = 13.7 Hz, 1 H), 4.64 (d, J = 13.7 Hz, 1 H), 3.83 (d, J = 15.0 Hz, 1 H), 4.52 (d, J = 15.0 Hz, 1 H), 3.45−3.30 (br, 1 H), 1.96 (s, 3 H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 140.9, 139.1 138.8, 136.6, 131.3, 131.1, 129.0, 128.7, 128.4, 128.2, 127.0, 126.1, 116.9, 89.4, 58.2, 36.2, 23.8; MS (EI) m/z (%) 318 (M^+, 18.72), 91 (100); IR (neat) 3395, 3025, 2913, 1642, 1600, 1493, 1451, 1371, 1264 cm^{-1}; HRMS (EI) calcd for $C_{22}H_{22}O_2$ (M^+) 318.1620. Found 318.1616.

2. Synthesis of 3,5-Dibenzyl-4-isopropyl-5,6-dihydropyran-2-one (12f). A solution of (*Z*)-3-benzyl-5-benzylidene-4-(prop-1-en-2-yl)-5,6-dihydropyran-2-one **3f** (47 mg, 0.15 mmol) and Pd/C (5 mg) in 1 mL of CH_3OH under 1 atm of H_2 atmosphere was stirred at room temperature for 5 h. After evaporation, the residue was purified by column chromatography on silica gel to afford 42 mg (88%) of the product **12f**: Liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.40−7.12 (m, 10 H), 4.14 (dd, J = 11.1, 1.2 Hz, 1 H), 4.02 (dd, J = 11.1, 0.9 Hz, 1 H), 3.91 (d, J = 15.0 Hz, 1 H), 3.75 (d, J = 15.0 Hz, 1 H), 3.25−3.10 (m, 1 H), 2.94 (dd, J = 13.2, 3.0 Hz, 1 H), 2.74 (dd, J = 14.1, 12.0 Hz, 1 H), 2.55−2.45 (m, 1 H), 1.25 (d, J = 6.6 Hz, 3 H), 1.10 (d, J = 6.9 Hz, 3 H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 165.8, 162.6, 139.6, 138.6, 129.1, 128.7, 128.5, 128.0, 126.8, 126.0, 124.7, 67.2, 36.2, 36.0, 31.9, 31.2, 21.7, 20.3; MS (EI) m/z (%) 320 (M^+, 28.87), 91 (100); IR (neat) 3061, 3027, 2966, 2935, 1711, 1602, 1494, 1467, 1453, 1130 cm^{-1}; HRMS (EI) calcd for $C_{22}H_{24}O_2$ (M^+) 320.1776. Found 320.1772.

3. Synthesis of (*Z*)-5-(Iodomethylene)-3-methyl-4-(prop-1-en-2-yl)-5,6-dihydropyran-2-one (13m). A solution of 3-methyl-5-methylene-4-(prop-1-en-2-yl)-5,6-dihydropyran-2-one **3m** (118 mg, 0.50 mmol) and *N*-iodosuccinimide (230 mg, 1.02 mmol) in 4 mL of dry CH_3CN was refluxed for 10 h. After evaporation, the residue was purified by column chromatography on silica gel to afford 128 mg (88%) of the product **13m**: Liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 6.75 (s, 1 H), 5.34 (s, 1 H), 5.01 (s, 2 H), 4.86 (s, 1 H), 1.89 (s, 3 H), 1.87 (s, 3 H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 165.1, 149.4, 138.8, 137.0, 121.1, 117.8, 84.4, 72.8, 22.2, 14.3; MS (EI) m/z (%) 290 (M^+, 1.98), 91 (100); IR (neat) 3068, 2919, 1717, 1643, 1600, 1452, 1395, 1255, 1129 cm^{-1}; HRMS (EI) calcd for $C_{10}H_{11}O_2I$ (M^+) 289.9804. Found 298.9809.

4. Synthesis of (*Z*)-5-(*m*-Methoxybenzylidene)-3-methyl-4-(prop-1-en-2-yl)-5,6-dihydropyran-2-one (14m). A mixture of **13m** (44 mg, 0.15 mmol), 3-methoxyphenylboronic acid (45 mg, 0.3 mmol), Pd(PPh_3)$_4$ (9 mg, 5 mol %), and $K_3PO_4\cdot 3H_2O$ (81 mg, 0.3 mmol) in toluene (2 mL) was refluxed under Ar for 10 min. After evaporation, the residue was purified by column chromatography on silica gel to afford **14m** (34 mg, 83%): Liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.30 (t, J = 8.0 Hz, 1 H), 6.86 (d, J = 8.0 Hz, 1 H), 6.77 (s, 1 H), 6.73 (d, J = 8.0 Hz, 1 H), 6.68 (s, 1 H), 5.38 (s, 1 H), 5.22 (s, 2 H), 4.93 (s, 1 H), 3.81 (s, 3 H), 2.00 (s, 3 H), 1.94 (s, 3 H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 165.3, 159.5, 152.7, 139.7, 136.4, 132.0, 129.5, 128.5, 121.8, 121.4, 117.5, 114.6, 113.7, 66.8, 55.2, 22.5, 14.7; MS (EI) m/z (%) 270 (M^+, 42.13), 226 (100); IR (neat) 2922, 1713, 1598, 1576, 1489, 1458, 1432, 1261, 1127 cm^{-1}; HRMS (EI) calcd for $C_{17}H_{18}O_3$ (M^+) 270.1256. Found 270.1261.

5. Synthesis of 3-Methyl-5-(*Z*)-(((*E*)-3-phenylvinyl)methylene)-4-(prop-1-en-2-yl)-5,6-dihydropyran-2-one (15m). A solution of **13m** (40 mg, 0.14 mmol), (*E*)-styrylboronic acid (41 mg, 0.28 mmol), Pd(PPh_3)$_4$ (8 mg, 5 mol %), and $K_3PO_4\cdot 3H_2O$ (76 mg, 0.28 mmol) in toluene (2 mL) was refluxed under Ar for 0.5 h. After evaporation, the residue was purified by column chromatography on silica gel to afford **15m** (52 mg, 100%): Liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.46−7.40 (m, 2 H), 7.40−7.20 (m, 3 H), 6.94 (dd, J = 15.2, 11.4 Hz, 1 H), 6.78 (d, J = 15.2 Hz, 1 H), 6.45 (d, J = 11.4 Hz, 1 H), 5.39−5.32 (m, 1 H), 5.22−5.18 (m, 2 H), 4.90−4.82 (m, 1 H), 1.98 (s, 3 H), 1.93−1.90 (m, 3 H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 165.8, 152.8, 139.5, 138.3, 136.4, 130.6, 128.8, 128.7, 127.0, 126.8, 122.3, 121.2, 117.1, 66.1, 22.6, 14.6; MS (EI) m/z (%) 266 (M^+, 6.70), 84 (100); IR (neat) 3033, 2922, 2855, 1706, 1616, 1582, 1449, 1258, 1128 cm^{-1}; HRMS (EI) calcd for $C_{18}H_{18}O_2$ (M^+) 266.1307. Found 266.1303.

6. Synthesis of (*Z*)-3-Methyl-5-(3-phenylprop-2-ynylidene)-4-(prop-1-en-2-yl)-5,6-dihydropyran-2-one (16m). A mixture of **13m** (39 mg, 0.14 mmol), phenylacetylene (32 mg, 0.3 mmol), $PdCl_2(PPh_3)_2$ (5 mg, 5 mol %), CuI (4 mg, 10 mol %), K_2CO_3 (42 mg, 0.3 mmol), and CH_3CN (2 mL) was stirred at rt under Ar for 1 h. Evaporation and purification by column chromatography on silica gel afforded **16m** (33 mg, 93%): Liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.50−7.40 (m, 2 H), 7.40−7.27 (m, 3 H), 5.96 (s, 1 H), 5.35 (s, 1 H), 5.29 (s, 2 H), 4.88 (s, 1 H), 1.98 (s, 3 H), 1.91 (s, 3 H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 165.1, 150.7, 138.8, 137.8, 131.5, 129.0, 128.5, 122.6, 122.4, 117.6, 109.6, 102.0, 85.4, 67.9, 22.3, 14.6; MS (EI) m/z (%) 264 (M^+, 46.71), 178 (100); IR (neat) 3080, 2919, 2854, 2187, 1709, 1642, 1598, 1489, 1442, 1256, 1130 cm^{-1}; HRMS (EI) calcd for $C_{18}H_{16}O_2$ (M^+) 264.1150. Found 264.1153.

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Supporting Information Available: All experimental procedures and analytical data for all starting materials and products; copies of ^{1}H/^{13}C NMR spectra of all new compounds; ORTEP drawing of **2d** and *epi*-**2d**; the CIF files of all crystals. This material is available free of charge via the Internet at http://pubs.acs.org.

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